Study Title: Interactions with large language models to promote positive attitudes toward refugees

1) Has any data been collected for this study already?
No, no data for the pre-registered confirmatory study has been collected yet.

2) What's the main question being asked or hypothesis being tested in this study?
The study aims to examine whether interactions with an intervention chatbot lead to more positive attitudes towards refugees using a within-subjects pre-post design. Each participant completes one baseline attitude measure before the intervention and one post-intervention measure afterward. Attitudes are a multi-dimensional construct. We distinguish three primary dependent variables and test each in a separate model, with one primary hypothesis per dimension.

Primary Hypotheses:

Participants will show higher post-intervention attitude scores compared to their own pre-intervention baseline.

- H1 (Perceptions): Participants will report higher perception scores after the intervention compared to their own pre-intervention baseline.
- H2 (Refugee support): Participants will report higher refugee support scores after the intervention compared to their own pre-intervention baseline.
- H3 (Policy endorsement): Participants will report higher policy endorsement scores after the intervention compared to their own pre-intervention baseline.

3) Describe the key dependent variable(s) specifying how they will be measured.
The primary dependent variables are participants' attitudes toward refugees, measured once before and once after the chatbot interactions.

We measure using a multi-item survey consisting of three dimensions:

- Perceptions: cognitive and affective evaluations of refugees, adapted from the asylum-seeker component of Hangartner et al. (2019).
- Refugee support: emotional concern and perceived responsibility toward refugees, drawing on items used by Czarnek et al. (2025).
- Policy endorsement: support for refugee-related policies, adapted from measures used in Vlasceanu et al. (2024).

Items are answered on Likert-type scales or 0-100 sliders. Each dimension will be analyzed separately. For each dimension, we hypothesize that participants' post-intervention scores will be higher than their pre-intervention scores.

4) How many and which conditions will participants be assigned to?
Participants will be randomly assigned to one of six conditions. This study is within-subjects only with respect to the pre-post comparison. Four of these conditions are the treatment groups, in which participants interact with different LLM-powered chatbot interventions.

Treatment groups:

- Factual Debunking Chatbot: addresses common misconceptions about refugees by providing fact-based responses.

- Self-Reflection Chatbot: guides participants through self-reflection and internal reasoning without introducing new factual information.
- Empathy Chatbot: designed to increase empathy toward refugees by encouraging perspective-taking, emotional engagement, and reflection on refugees' experiences.
- Roleplay Chatbot: designed to bring participants into the perspective of a refugee through an empathetic, first-person conversational roleplay.

The other two conditions serve as control groups.

Control group:

- Neutral Chatbot: guides a conversation unrelated to refugees, which thus controls for the general experience of engaging with a chatbot.

Additional comparison group:

- Static Information Condition: a written, non-interactive summary about refugees, controlling for the interaction format (i.e., exposure to information without interaction).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will primarily analyze attitudes using linear mixed- effects models (LMMs). The neutral-bot condition will serve as the reference category. Confirmatory tests will be conducted using Holm-corrected pairwise comparisons ($\alpha = 0.05$). Exploratory analysis will consist of additional contrasts within the same LMM framework.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Participants will be excluded from the analysis if they fail attention checks, which are embedded within the surveys, complete the study unusually quickly (less than one-third of the median completion time), or provide incoherent or low-effort responses, such as very short answers (fewer than three words) or placeholder/dummy text. Additional exclusions will apply to participants who encounter technical issues during their interaction with the LLM-powered chatbot. The number of participants excluded under each criterion, along with the justification for their exclusion, will be reported in the results to ensure transparency and data integrity.

7) How many observations will be collected or what will determine sample size?

An a priori power analysis via G*Power (version 3.1.9.7) using a t-test (test family: t tests; statistical test: Means – difference between two dependent means, matched pairs) for the confirmatory pre-post comparisons (effect size d = 0.35, holm-corrected significance level 0.0125, power 0.80) indicates a sample size of 486 participants (81 per group). To account for potential attrition, the sample size will be increased by 20%, resulting in an intended total of approximately 583 participants.

Participants will be recruited through Prolific, targeting a diverse sample of adults residing in the United Kingdom. Inclusion criteria are: minimum age of 18, English as a first language, an approval rate of at least 98%, and at least 50 previous submissions.

8) Anything else you would like to pre-register?

In addition to the primary analyses, several elements of the study will be documented for transparency. We will collect standard sociodemographic variables. These will not be used in confirmatory analyses but may be examined descriptively or in exploratory subgroup analyses.

9) References

Czarnek, G., Orchinik, L., Rand, D. G., Lin, H., Xu, H. G., Costello, T., Pennycook, G., & Rand, D. G. (2025). Addressing climate change skepticism and inaction using human-AI dialogues [Working paper]. OSF. https://osf.io/preprints/psyarxiv/mqcwj_v1

Hangartner, D., Dinas, E., Marbach, M., Matakos, K., & Xefteris, D. (2019). Does exposure to the refugee crisis make natives more hostile? American Political Science Review, 113(2), 442–455. https://doi.org/10.1017/S0003055418000813

Vlasceanu, M., Doell, K. C., Bak-Coleman, J. B., et al. (2024). Addressing climate change with behavioral science: A global intervention tournament in 63 countries. Science Advances, 10(6), eadj5778. https://doi.org/10.1126/sciadv.adj5778